



Regional Variations in Burden of Chronic Obstructive Pulmonary Disease

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OBJECTIVES

To examine the extent of **heterogeneity in burden of COPD** across geographic regions in British Columbia



To examine the **heterogeneity in trends of COPD burden** across geographic regions

METHODS

Design

- Retrospective Longitudinal population - based cohort study

Data

- Population Data BC
- Vital Statistics
- Consolidation files
- Hospital Separations
- Medical Services Plan

Time

- 2010-2020

Variable of Interest

- Health Services Delivery Area

Outcome

- Prevalence
- Incidence
- Mortality
- Hospitalization

Statistical Analysis

- Descriptive Statistics
- Adjusted Generalized Linear Model
- Trend Adjusted Generalized Linear Model
- $P < 0.05$ (Significance level)

RESULTS



Patients

312,014



Female

48%



Mean Age

68.2 years

Period Prevalence (%)

6.8

Cumulative Incidence
(per 100,000 PY at risk)

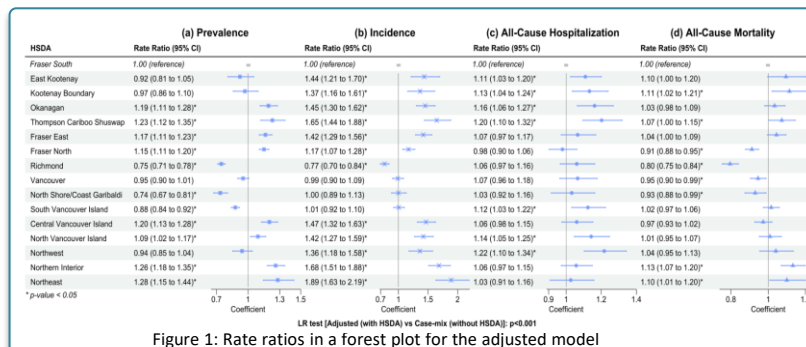
5.7

Cumulative Hospitalization
(per 100,000 COPD PY)

73136.1

Cumulative Mortality
(per 100,000 COPD PY)

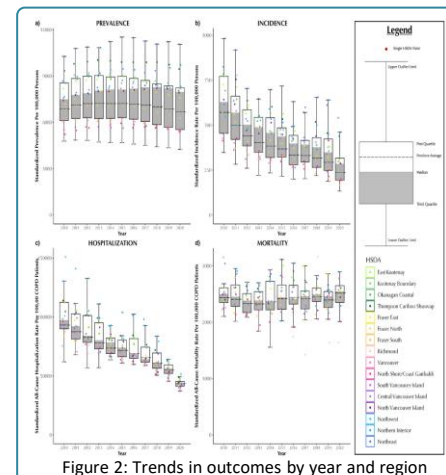
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There are significant differences in COPD outcomes based on location

On average, there was up to a **three-fold** difference in **incidence and prevalence** rates and a **two-fold** difference in **mortality and hospitalization** rates across regions

Heterogeneity was more prominent in incidence and prevalence, compared with hospitalization and mortality.



The study emphasizes the importance of identifying and targeting system-level factors that affect the quality of care and outcomes of COPD, alongside our quest for finding better treatments